

- Application and database run as **Docker containers** for consistent and portable environments.

Advantages of the Proposed System:

- Eliminates manual errors and delays in slot allocation.
- Ensures security through token-based authentication and role management.
- Reduces congestion by providing real-time slot availability.
- Improves user satisfaction with a post-parking payment model.
- Scalable and adaptable for future IoT or mobile integrations.

❖ **Module Specification:** [No. of Modules & their Specification]

The “**First Park Then Pay**” application is organized into **four main modules**, each focusing on a specific functionality within the system. It follows a **SaaS (Software as a Service)** model where each customer (parking provider) can create an account and manage their own parking setup, charges, and billing through a secure web application.

1. User Management Module

Purpose:

Manages all user accounts, authentication, and authorization through **Key cloak**.

Specifications:

- User authentication and authorization handled by **Key cloak**.
- Role-based access for **Customer (Parking Owner)** and **Admin**.
- Each customer can register and maintain their own parking business account.
- Secure login, token-based access, and session handling.
- Profile details and credentials synced from Key cloak.

2. Vehicle Management Module

Purpose:

Maintains vehicle type configurations and associates parking slots with those types.

Specifications:

- Customers can create and manage vehicle types (e.g., Car, Bike, Van, EV, etc.).
- Each vehicle type includes configuration details such as:
 - **Block Name**
 - **Slot Number**
 - **Per-Hour Parking Charges**
- Vehicle type data is used for billing and parking allocation.
- Dropdown-based vehicle type selection ensures consistency.
- Allows expansion for new vehicle types as needed.

3. Billing and Payment Module

Purpose:

Handles billing, payment tracking, and transaction verification for parked vehicles.

Specifications:

- Automatically calculates charges based on **parking duration** and **vehicle type**.
- Payment mode options: **Cash** or **Online**.
- Validates whether the bill is paid or pending before releasing the slot.
- Maintains detailed payment history for both admin and users.
- Integrated with vehicle data for accurate charge mapping.

4. Dashboard Module

Purpose:

Acts as the central interface for both users and admins to monitor system activities and performance.

Specifications:

- **User Dashboard:** Displays registered vehicle types, created blocks/slots, and transaction history.
- **Admin Dashboard:** Summarizes customer accounts, total payments, and parking activities.
- Real-time statistics such as total vehicles parked, earnings, and slot usage.
- Data visualizations through charts or tables for better analysis.